University of Waterloo
School of Computer Science
CS 486/686, **SAMPLE** Final Examination
Winter 2025

- **Instructor: Jesse Hoey and Victor Zhong**
- **Date: Sometime Before April 22nd, 9:00am**
- **Location: Wherever**
- **Time: Whenever**
- **There are 5 questions on this exam**
- **There are 6 pages in this exam**
- **There are a total of 100 marks on the exam**
- **You have 150 minutes to complete the exam**
- **Non-programmable calculators are allowed**
- **Write your answers on a separate sheet. On the REAL final, you will write directly on the exam (space will be provided)**

Question 1: Short Answers (10 points out of a total of 100 points)

(1a) [**2 points**] Given this set of data samples drawn from some Bayesian network over 4 binary variables Z, X, Y, W, what is the probability that W is true (t) given X is true (t) and Y is false (f) if using maximum likelihood learning of a binomial distribution.

Z	X	Y	W
t	t	f	f
t	f	t	f
t	t	f	t
t	t	t	t
t	t	f	f
t	f	f	t
t	f	t	t

(1b) [**2 points**] A heuristic for a search problem is an estimate of the true cost to the goal. Is the following statement true or false: *an admissible heuristic is always greater than the true cost.*

(1c) [**2 points**] True or false: it is OK to ignore data with missing features/attributes when learning, but only if the data is *missing at random*

(1d) [**2 points**] Complete the following sentence: A* search uses a priority queue of nodes ranked by ______________, and so is a mixture of ______________ and best-first search.

(1e) [**2 points**] True or False: Overfitting in maching learning refers to when your model has too much/little bias.

Question 2: Maximum Likelihood Learning (15 points out of a total of 100 points)

A *mixture of Gaussians* model is a Bayesian network $C \rightarrow X$ where C is a binary variable with values $\{1, 2\}$ and a binomial distribution $P(C) = \{\theta_c, 1 - \theta_c\}$ and X is a continuous variable with a Gaussian probability distribution:

$$P(x|c) = \frac{1}{\sqrt{2\pi}\sigma_c} e^{\frac{-(x-\mu_c)^2}{2\sigma_c^2}}$$

the parameters of this model are thus $\Theta = \{\theta_c, \mu_1, \sigma_1, \mu_2, \sigma_2\}$ where μ_i is the mean of the Gaussian (normal) distribution for category $c = i$ and σ_i is the variance of the Gaussian distribution for category $c = i$. You are given a dataset $\mathbf{d} = \{c_1, x_1, c_2, x_2, \ldots, c_M, x_M\}$ giving complete data - one value for X and one value for C for each datapoint.

In this question, you will derive equations for the maximum likelihood estimates of the five parameters. Recall that this involves solving the following equation

$$\theta_{ML} = \arg\max_{\theta} P(\mathbf{d}|\Theta)$$

where $\theta \in \Theta$ is any parameter and θ_{ML} is the maximum likelihood estimate of that parameter. One does this by first writing down $P(\mathbf{d}|\Theta)$ using the binomial distribution $\theta_c^{c_j}(1-\theta_c)^{c_j}$ and normal distribution $P(x_j|c_j)$ given above for data point j.

You may need the following derivatives

$$\frac{\partial}{\partial x} e^{f(x)} = e^{f(x)} \frac{\partial}{\partial x} f(x)$$

$$\frac{\partial}{\partial x}\left(\frac{1}{f(x)^2}\right) = -\frac{2}{f(x)^3}\frac{\partial}{\partial x} f(x)$$

$$\frac{\partial}{\partial x} log(f(x)) = \frac{1}{f(x)}\frac{\partial}{\partial x} f(x)$$

(1a) [**5 points**] Derive an equation for the maximum likelihood estimate of θ_C.

(1b) [**5 points**] Derive an equation for the maximum likelihood estimate of μ_i.

(1c) [**5 points**] Derive an equation for the maximum likelihood estimate of σ_i.

Question 3: Decision Trees (20 points out of a total of 100 points)

You are trying to predict who will win the Stupley Can™, annual prize handed out to the winner of the National Chokey League™ (Chokey is a team sport that is popular in Olertawo). There are 12 teams in the league and you have a database of historical information about previous winners and losers. Your database is a table where each row indicates a team's performance. The statistics you are given include the team's win percentage (binary variable $\in \{hi, lo\}$), whether they got more than 100 minutes of penalties/season on average $\in \{true, false\}$ and if the average number of goals they scored was greater than the team average for the league $\in \{true, false\}$. Finally, the last column gives the number of times this team has won the Stupley Can ($\in \{1, 2, 3\}$). Here is the dataset:

`win %`	`>100mins penalties`	`>avg goals`	`# Stupley Can wins`
lo	false	yes	3
hi	true	no	2
hi	true	yes	3
lo	false	no	1
lo	true	no	1
hi	true	no	1
hi	false	yes	3
lo	false	yes	3
hi	true	no	2
lo	true	no	2
lo	true	yes	3
lo	false	no	1

Recall that, given a set E of N training examples, if the number of examples with output (target) feature $Y = y_i$ is N_i, then

$$P(Y = y_i) = P(y_i) = \frac{N_i}{N}$$

and the total information content for the set E is:

$$I(E) = -\sum_{y_i \in Y} P(y_i) \log_2 P(y_i)$$

Using information gain (weighted by fractions of documents) as your feature selection mechanism, construct a decision tree that predicts how many times a team will win the Stupley Can based on the features of `win%`, `penalties` and `goals`. Continue splitting until no more attributes are available, or until all data agrees on the target attribute (`# of Stupley Can wins`). Break ties in feature selection arbitrarily. Use the majority class at each leaf as your point estimate (prediction). In case of a tie, choose the lower wins number. Show your work and draw a graph of your final decision tree.

Question 4: Neural Networks (30 points out of a total of 100 points)

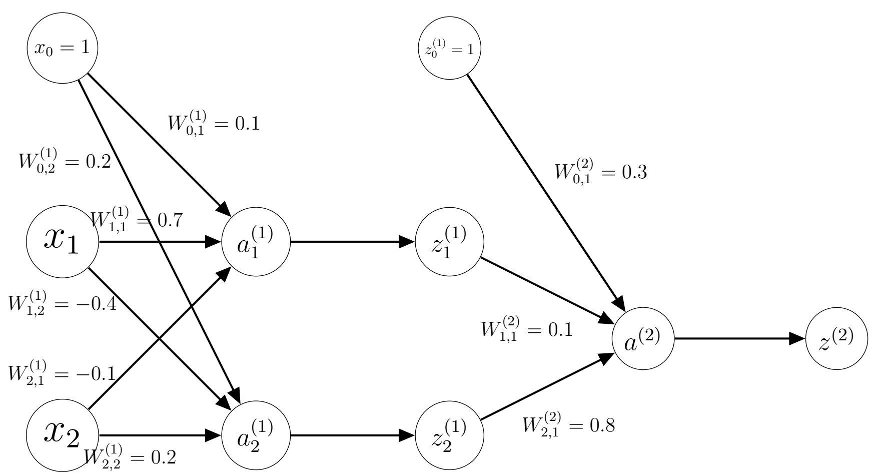
Carolyn is an machine learning engineer working with a soccer team. Carolyn would like to predict whether the team will win their next game. She therefore treats this as a binary classification problem. After querying the team's records, Carolyn assembles a dataset consisting of previous games. Each example consists of features regarding the game, and the target is whether or not her team won the game. The features and target are:

Features:

- x_1: The game number (out of the total games for a season)
- x_2: The number of injured players

Target: Whether the team won or lost the game.

Carolyn trains the following neural network, using a sigmoid $f(x) = \frac{1}{1+e^{-x}}$ as the activation function for both hidden layers. The network is trained to minimize the squared error function (for each datapoint): $E(\hat{y}, y) = (y - \hat{y})^2$



Recall that, starting with l, the error is computed for the output layer (l) as:

$$\vec{\delta}^{(l)} = \frac{\partial Error}{\partial \vec{z}^{(l)}} \times f'(\vec{a}^{(l)}) \tag{1}$$

and the error for the other layers starting from $l-1$ (next to last layer):

$$\vec{\delta}^{(l)} = \left(\vec{\delta}^{(l+1)} \cdot \mathbf{W}^{(l+1)T}\right) \times f'(\vec{a}^{(l)}) \tag{2}$$

then the weight updates for the input layer are:

$$\frac{\partial Error}{\partial \mathbf{W}^{(l)}} = \vec{x}^T \vec{\delta}^{(l)} \tag{3}$$

and other layers:

$$\frac{\partial Error}{\partial \mathbf{W}^{(l)}} = \vec{z}^{(l-1)T}\vec{\delta}^{(l)} \tag{4}$$

and finally update the weights are updated using:

$$\mathbf{W}^{(l)} \leftarrow \mathbf{W}^{(l)} - \eta \frac{\partial Error}{\partial \mathbf{W}^{(l)}}$$

3(a) [**8 points**] Execute the forward pass of this neural network for the example $x = \langle 1, 2 \rangle$. Use the weights that are labelled on the diagram above. The final output should be the network's prediction, $\hat{y} = z^{(2)}$.

3(b) [**2 points**] Suppose that the target for example $x = \langle 1, 2 \rangle$ is $y = 1$ (game was won). What is the value of the error function, given the network's prediction from part (a)?

3(c) [**17 points**] Calculate the partial derivative of the error with respect to $W^{(2)}_{1,1}$ (i.e. $\frac{\partial E}{\partial W^{(2)}_{1,1}}$). Be sure to show all of your steps.

3(d) [**3 points**] Using the partial derivative you calculated in part (a), update the value of the weight $W^{(2)}_{1,1}$. Use a learning rate of $\eta = 0.1$.

Question 5: Bayesian Networks (25 points out of a total of 100 points)

A binary sensor (S) is used to detect overheating of the core in a nuclear reactor. The sensor may be working properly (*S_ok=true*) or malfunctioning (*S_ok=false*). If the reactor core overheats (C = *true*), and the sensor is working properly, the sensor will read *S=true* with certainty (probability *1.0*). If the core doesn't overheat, and the sensor is working properly, the sensor reads *S=true* 3% of the time (false positives). If the sensor is malfunctioning then the sensor will read *S=true* with probability *0.5* regardless of the state of the core. A positive reading on the sensor (*S=true*) causes an alarm to go off (*A=true*) 95% of the time, and causes all the doors in the reactor to lock (*L=true*) 99% of the time. If *S=false*, the alarm never goes off, but the doors lock with probability 0.02. The sensor malfunctions with probability 0.01, and the reactor core has a 10% chance of overheating.

(3a) [**5 points**] Draw a Bayesian network representing the knowledge you have about the nuclear reactor above.

(3b) [**10 points**] Write down a complete set of factors representing all the knowledge about the conditional probabilities in the BN. You should have 5 factors.

Answer the following queries about the nuclear reactor using your Bayesian network and Variable Elimination. In the following questions, you can leave your answer as a product and/or sum of the real numbers found in your BN (i.e. you don't need to explicitly do the computations on your calculator, but the final answer should be in a form that could be put into a calculator, without any variables and Σ symbols).

(3c) [**2 points**] What is the probability of the reactor core overheating (given no evidence)?

(3d) [**8 points**] What is the probability that the reactor core is overheating given that the alarm is going off?